7.12 C-11 GUIDANCE TRANSMITTER UNIT (GTU) CHECK

INITIAL SETUP:

Tools and Special Tools

References
020-00089 TI Manual Software Tools

Materials/Parts

Equipment Condition
External Electrical Power Available (720-00015 Kubota GL11000)

Personnel Required
Technician (2)

GENERAL INFORMATION

The following procedure ensures that the GTU is configured properly.

GTU Transmitter Check Procedure

Uplink Monitor

NOTE

- See Figure 7-1 GTU Test Interface for the remainder of this section.
- Sign of DDM values indicated on PIR is opposite values indicated in GTU Test tool.
- GTU transmit calibration uses sign indicated on PIR for values measured and entered in GtuTxCal tool.
- In GTU Test tool, “Use transmit calibration” checkbox must be checked when verifying transmit calibration and performing a monitor calibration.

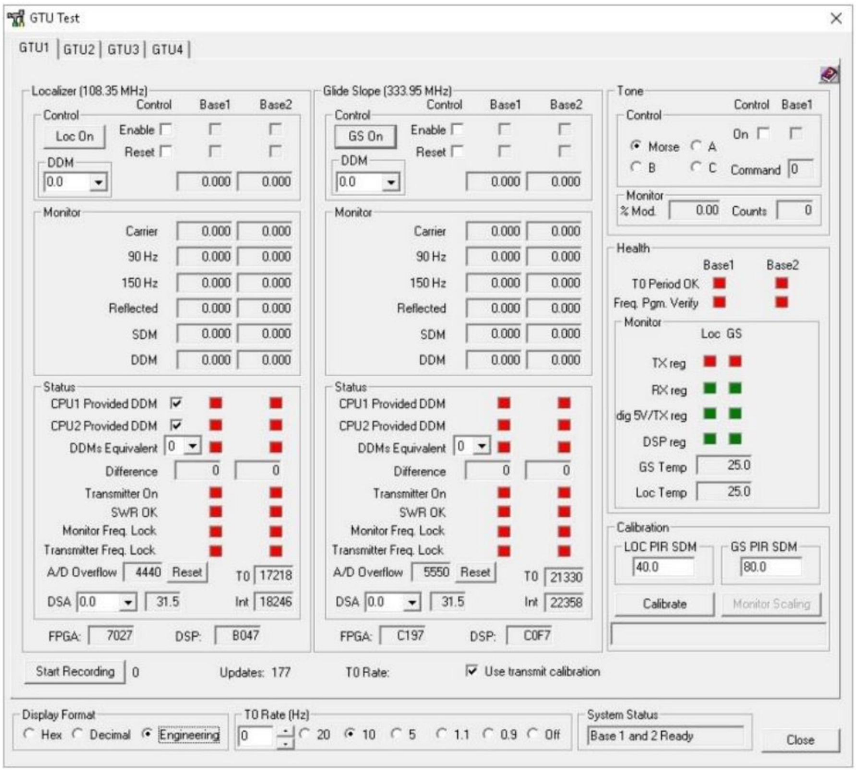

Figure 7-1. GTU Test Interface

1. Assemble PIR and position it in front of uplink antenna array at a distance of 30 feet from base of uplink tower.
2. Enable both the LOC and GS transmitters via GTU Test Interface.
3. Ensure carrier power to be between 2100 and 2300 mV for both localizer and glideslope.
4. If they are out of tolerance, complete following:
    - To accomplish this tuning, read carrier power level for localizer and, if it is below 2200mV, then reduce DSA attenuation value.
    - If carrier is above 2200mV, then increase DSA attenuation value.
    - Repeat this process for glide slope.
    - Record DSA and carrier amplitude on test datasheet for both localizer and glideslope
5. Allow 15 minutes transmitter ON time prior to continuing evaluation/test.
6. If GTU has already achieved normal operating temperature and has been on for 15 minutes, continue to next step.
7. Power on PIR and ensure PIR frequency select knobs are selected to frequency of localizer transmitter.
8. Record both Monitor Carrier levels from GTU Test window and PIR SDM readings for localizer and glideslope

9. Input LOC PIR SDM and GS PIR SDM as measured above in Calibration section of GTU Test Interface.
10. Depress “Calibrate” button.
11. When calibration is complete, System Status window will display “Calibration Complete, Run Monitor Scaling.”
12. Depress “Monitor Scaling” button in Calibration section of GTU Test Interface.
13. Click on calculate buttons for both localizer and Glideslope sides.
14. Verify calculated values transfer down to “Calculated Scaling” windows, in Monitor Scaling window.
15. Click on “Send to Bases” button.
16. Verify Base1 and Base2 read-backs are same as control fields. (The Offset numbers will be rounded to closest integer).
17. Depress “Close” button in Monitor Scaling window.
18. Record successful GTU calibration.

NOTE

- SDM values for localizer shall be 40±1%, and DDM values shall be setting ±.002.
- PIR readings on DDM has reversed polarity.

19. For localizer signal, change DDM control settings per list below and record monitor SDM, monitor DDM, PIR SDM, and PIR DDM values.
    - **+.20, +.18, +.15, +.12, +.09, +.06, +.03, 0, -.03, -.06, -.09, -.12, -.15, -.18, -.20**
20. Record validation in Appendix F.

NOTE

- SDM values for glide slope shall be 80±2%, and DDM values shall be setting ±.004.

21. For glide slope signal, change DDM control settings per list below and record monitor SDM, Monitor DDM, PIR SDM, and PIR DDM values.
    - **+.24, +.22, +.2, +.18, +.15, +.12, +.09, +.06, +.03, 0, -.03, -.06, -.09, -.12, -.15, -.18, -.2, -.22, -.24**
22. Record validation in Appendix F
23. Record GTU monitor values as listed in Appendix F.
24. On GTU test tool, click on “Close” to exit test tool.

TLS Uplink Field Strength Measurements

NOTE

Flag stake identifies position of TinySA operator when taking measurements.

1. Measure 50 feet in front of uplink tower and mark ground with a wire flag stake.
2. Assemble TinySA ensuring 27-inch antenna attached to low port of TinySA and antenna fully extended.
3. Power on TinySA and set center frequency to localizer frequency with a span of 100 KHz.
4. On TinySA menu, select Display-> Sweep Accuracy - > Precise, select Frequency - > RBW -> 600 KHz.
5. At TLS MIU, select GTU Test SW tool and enable localizer transmission.

NOTE

Use strongest value with sweep set to precise.

6. Walk to stake 50 feet in front of TLS uplink hold TinySA 5 feet above ground level with antenna positioned horizontally and measure localizer amplitude.
7. Use table below to determine allowed tolerance.
8. Install 11-inch antenna on TinySA low port and fully extend antenna.
9. Repeat above steps 3 through 7 with glide slope enabled on GTU test tool and tune TinySA to glide slope frequency

Antenna type	Description	Number of GTU's	Minimum Field Strength dBm
920-00772	Loc Normal Gain	1	-8
920-00771	GS Normal Gain	1	-6
920-01037-100	Loc High Gain	1	-14
920-00754	GS High Gain	1	-11
920-01037-100	Loc High Gain	4	-17
920-00754	GS High Gain	4	-9

Table 7-1.

Guidance Transmitter Check Documentation

1. Complete, sign, and file on-site TLS Technical Performance Record for GTU C-11.
2. Record completion of C-11 in Facility Maintenance Logbook.

END OF TASK

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